

UNIT – 3
ASSESSMENT – 1
OUTPUT

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1.

```
===== RESTART: C:/Users/Niharika Preejesh/OneDrive/  
SMART MEDICAL DIAGNOSIS SYSTEM
```

```
Initial Facts:  
- breathlessness  
- cough  
- fever
```

```
===== FORWARD CHAINING =====
```

```
Rule Fired:  
cough AND fever -> flu  
Derived Fact: flu
```

```
Rule Fired:  
breathlessness AND cough -> pneumonia  
Derived Fact: pneumonia
```

```
All Derived Facts:  
- pneumonia  
- fever  
- breathlessness  
- cough  
- flu
```

```
===== BACKWARD CHAINING =====
```

```
Checking Goal: pneumonia  
Using Rule: breathlessness AND cough -> pneumonia
```

```
Checking Goal: breathlessness  
breathlessness is already a fact.
```

```
Checking Goal: cough  
cough is already a fact.
```

```
Conclusion:  
Patient A has Pneumonia.
```

```
|
```

2.

```
===== RESTART: C:/Users/Niharika Preejesh/OneDrive/Document
===== AUTONOMOUS TRAFFIC MANAGEMENT =====

Initial Facts:
Vehicle(CarA)
Vehicle(Ambulance)
Behind(CarA,Ambulance)
EmergencyType(Ambulance)

--- Rule 1 ---
Vehicle(x) AND EmergencyType(x) -> ClearPath(x)

Unification:
x = Ambulance
MGU = {x/Ambulance}
Derived: ClearPath(Ambulance)

--- Rule 2 ---
ClearPath(x) -> GreenSignal(x)
Derived: GreenSignal(Ambulance)

--- Rule 3 ---
Vehicle(x) AND Behind(x,y) AND GreenSignal(y) -> Proceed(x)
Derived: Proceed(CarA)

===== RESOLUTION =====
Goal: Proceed(CarA)
Negated Goal: NOT Proceed(CarA)

Resolution:
Proceed(CarA)
NOT Proceed(CarA)
-----
EMPTY CLAUSE (□)

Conclusion:
Proceed(CarA) is PROVED.
|
```

3.

```
===== RESTART: C:/Users/Niharika Preejesh/OneDrive.
=====
AGRICULTURAL EXPERT REASONING SYSTEM
=====

Initial Facts:
CropWheat
SoilDry

Rule 1:
SoilDry -> IrrigationNeeded
Derived: IrrigationNeeded

Rule 2:
IrrigationNeeded AND CropWheat -> ApplyDripMethod
Derived: ApplyDripMethod

Goal: ApplyDripMethod
Negated Goal: NOT ApplyDripMethod

===== RESOLUTION REFUTATION =====

Derived Clause:
ApplyDripMethod

Negated Goal:
NOT ApplyDripMethod

Resolving:
ApplyDripMethod
NOT ApplyDripMethod
-----
EMPTY CLAUSE (□)

Conclusion:
ApplyDripMethod is PROVED.

===== CROP RISK ANALYSIS =====
CropAtRisk cannot be concluded from the given facts.
```